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Towards a mechanistic understanding of insect life history evolution: oxygen-dependent induction of moulting explains moulting sizes

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Moults characterise insect growth trajectories, typically following a consistent pattern known as Dyar's rule; proportional size increments remain constant across inter-instar moults. Empirical work suggests that oxygen limitation triggers moulting. The insect respiratory system, and its oxygen supply capacity, grows primarily at moults. It is hypothesized that the oxygen demand increases with increasing body mass, eventually meeting the oxygen supply capacity at an instar-specific critical mass where moulting is triggered. Deriving from this hypothesis, we develop a novel mathematical model for moulting and growth in insect larvae. Our mechanistic model has great success in predicting moulting sizes in four butterfly species, indirectly supporting a size-dependent mechanism underlying moulting. The results demonstrate that an oxygen-dependent induction of moulting mechanism would be sufficient to explain moulting sizes in the study species. Model predictions deviated slightly from Dyar's rule, the deviations being typically negligible within the present data range. The developmental decisions (e.g. moulting) made by growing larvae significantly affect age and size at maturity, which has important life history implications. The successful modelling of moulting presented here provides a novel framework for the development of realistic insect growth models, which are required for a better understanding of life history evolution. © 2015 The Linnean Society of London, *Biological Journal of the Linnean Society*, 2016, **117**, 586–600.

ADDITIONAL KEYWORDS: development time – growth rate – growth trajectory – larval instars – mechanistic growth model.

INTRODUCTION

The larval stage of holometabolous insects is specialized for growth and is ecologically very different from the adult stage. During the larval stage, many insects accumulate resources that ultimately will be converted into reproduction in the adult stage, and a proper understanding of larval growth is therefore central for life history theory. Moulting between larval instars is a fundamental characteristic of insect growth, which makes the growth trajectory discontinuous (Chapman, 1998). The size of inextensible structures, such as mouthparts, increases only at moults, whereas body mass increases between moults during each larval instar. During intermoult growth, the absolute mass gain per unit time increases towards late instars (e.g. Nylin, Gotthard & Wiklund, 1996; Nijhout, Davidowitz & Roff, 2006; Tammaru & Esperk, 2007; Grunert *et al.*, 2015). For a given insect species the proportional size increment (growth ratio) of linear dimensions of the exoskeleton tends to remain constant throughout all larva-to-larva moults, which is an empirically broadly supported pattern known as Dyar's rule (Dyar, 1890; Hutchinson *et al.*, 1997; Berg & Merritt, 2009). Although different mechanisms such as epicuticle

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folding or the numbers of cell divisions at moults have been suggested to underlie Dyar's rule and interspecific variation in the proportional size increments at moults (reviewed by Hutchinson *et al.*, 1997), the mechanistic basis of the rule still remains unclear.

Ultimately, there must be several constraints on larval growth that necessitate moulting at some point to maintain growth efficiency. To investigate this issue, Hutchinson *et al.* (1997) developed an 'investment principle' – investment of accumulated reserves in building larger food acquisition structures (e.g. mouthparts) at a moult. A central assumption was that growth is limited by food acquisition rate, which in turn depends on the size of inextensible structures that grow only at moults. Occasional growth of these structures is necessary for the maintenance of high growth rate through the larval stage. An optimization analysis of the timing of moults and size increments at them showed that the 'investment principle' can provide an adaptive ultimate explanation for the empirically observed pattern of Dyar's rule, and for the moult-by-moult decrease in the growth ratio, which is the most common deviation from the rule (Hutchinson *et al.*, 1997). Thus, the 'investment principle' provides ultimate explanations for moulting strategies, but it does not provide a proximate mechanism for moult induction. For instance, food ingestion rate does not constrain growth before moulting in Lepidoptera (Esperk & Tammaru, 2004), which means that constraints on growth due to food ingestion rate cannot induce moulting. Understanding the proximate basis of moulting is necessary for understanding and analysing phenotypic plasticity in moulting, and the consequent implications for life history evolution (cf. Flatt & Heyland, 2011). Furthermore, the evolution of moulting strategies may also be constrained by the proximate basis of moult induction. Owing to these reasons, a mechanistic model of moult induction is required, but such a model is currently missing.

It has been suggested that moulting is induced by limited oxygen supply for growing tissues because the tracheal respiratory system, and its oxygen supply capacity, grows primarily at moults (Greenberg & Ar, 1996; Greenlee & Harrison, 2004a, 2005; Callier & Nijhout, 2011). Even though there may be some within-instar tracheal growth, which has been observed in the final instar but not in earlier instar larvae of the lepidopteran *Manduca sexta* (Callier & Nijhout, 2011; Helm & Davidowitz, 2013), it seems evident that the oxygen supply capacity remains relatively fixed between moults. Accordingly, the minimum oxygen partial pressure that allows normal metabolism and feeding increases towards the end of each larval instar both in *M. sexta* and in the orthopteran *Schistocerca americana*, suggesting that oxygen delivery to growing tissues would become compromised at some point (Greenlee & Harrison, 2004a, 2005). Moreover, in *M. sexta*, whole-organism oxygen consumption of larvae increases with growth (i.e. mass accumulation) early in each instar, but levels off (i.e. becomes independent of mass) late in each instar (Callier & Nijhout, 2011; but see Greenlee & Harrison, 2005 for a different pattern in CO_2 emission). This situation is consistent with a large oxygen safety margin (i.e. excess oxygen supply capacity) at the beginning of an instar, the safety margin diminishing with growth, and eventually disappearing completely at the end of the instar. Consequently, the tracheal system is assumed to function at its maximum capacity at the end of each instar, potentially being associated with oxygen limitation, but there may also be coincident changes in metabolism affecting oxygen need and consumption (Callier & Nijhout, 2012, 2014; see also Greenlee & Harrison, 2004a). Callier & Nijhout (2011) suggested that moulting is induced at the point (critical mass) at which whole-organism oxygen need meets the tracheal system supply capacity, that is, at the point when the oxygen safety margin disappears and (whole-organism) oxygen consumption becomes mass-independent. Finally, the number of moults increases with decreasing oxygen partial pressure in the coleopteran *Tenebrio molitor* (Greenberg & Ar, 1996), as would be expected if moulting is induced at the critical mass defined above and there is a threshold size for metamorphosis (Nijhout, 1975). This situation is because moult is induced at a lower (critical) mass with decreasing oxygen partial pressure. We refer to this proposed mechanism of moult induction as the oxygen-dependent induction of moulting (ODIM) hypothesis.

The size-dependent mechanism of moult induction stated by the ODIM hypothesis, however, holds only under benign growth conditions. Starved larvae are able to moult below a traditional empirical estimate for the critical mass (i.e. minimum mass where the time until pupation is not affected by nutrition; Nijhout & Williams, 1974; Nijhout *et al.*, 2006; Callier & Nijhout, 2011; Callier *et al.*, 2013) by using a size-independent mechanism of moult induction (Callier & Nijhout, 2011, 2013; Grunert *et al.*, 2015). Still, the ODIM hypothesis seems to be a plausible candidate for explaining the size-dependent mechanism for moulting under typical growth conditions (Callier & Nijhout, 2011).

Here, we derive from the ODIM hypothesis a mechanistic and mathematical description of insect growth and moulting. We fit the model to empirical data on growth trajectories, including information on

moulting sizes, of four butterflies to study if the mechanism stated by the ODIM hypothesis is sufficient to explain moulting sizes in the study species. In this way, we indirectly test the ODIM hypothesis. We also analyse how the predictions of the ODIM hypothesis relate to Dyar's rule (within-instar mass increments may follow Dyar's rule as mass generally scales to the cube of linear dimensions). In order to compare the data-derived predictions of the ODIM hypothesis to Dyar's rule, we construct an alternative model that is based on Dyar's rule for a reference.

MATERIAL AND METHODS

Mechanistic growth model

Details of model derivation are presented in Appendix A and the parameters are summarised in Table 1. Empirical data suggest that larval growth is exponential within each instar (e.g. Nylin *et al.*, 1996; Nijhout *et al.*, 2006). However, instantaneous growth rate decelerates at the end of instars (Esperk & Tammaru, 2004; Grunert *et al.*, 2015), which is known to take place after the attainment of the critical mass in the final instar *M. sexta* (Nijhout *et al.*, 2006). Hence, for each larval instar i ($i = 1, \ldots, n$), we assume exponential growth before the attainment of the critical mass, $m_{\mathrm{crit}}(i)$, and decelerating growth thereafter. Growth before the attainment of the critical mass is described by equation

$$m_{\mathrm{pre-crit}}(i,t) = m_0(i)e^{k(i)t}, \tag{1}$$

where $m_{\text{pre-crit}}(i, t)$ is mass (mg) of instar i at time t ($0 \le t \le t_{\mathrm{crit}}(i)$, where $t_{\mathrm{crit}}(i)$ is the time (days) from the beginning of growth in instar i to the attainment of critical mass of instar (i), $m_0(i)$ is the initial mass of instar i, and $k(i)$ is an instar-specific growth coefficient. In the absence of quantitative information about how growth decelerates, we arbitrarily specify growth after the attainment of the critical mass with the following function

$$m_{\mathrm{post-crit}}(i,t') = m_0(i)\exp\left[k(i)(t_{\mathrm{crit}}(i) + t' - \frac{t'^2}{2t_{\mathrm{post-crit}}})\right] \tag{2}$$

where $m_{\text{post-crit}}(i, t')$ is mass of instar i at time t', and $0 \le t' \le t_{\text{post-crit}}$, $t_{\text{post-crit}}$ being the time from the attainment of critical mass to moulting (or to the attainment of the peak larval mass at the final instar). Development rate, r, of the physiological preparation for moulting (or pupation), triggered by the attainment of the critical mass, determines $t_{\text{post-crit}}$ (see below). The instar-specific growth rate coefficient, $k(i)$, decreases instar by instar (Nijhout *et al.*, 2006), but as

Table 1. Summary of the parameters of the mechanistic growth model

Parameter type	Parameter	Description	Units	Value
General	α	Decrease rate of instar-specific growth rate coefficients [$k(i)$] in relation to instar initial mass	mg^{-1}	0.01
	σ	Ratio of early and late critical oxygen partial pressures for growth divided by the relative instar-wise decrease in maximum mass-specific tracheal conductance	Unit-less	2/3
Individual-specific (measured)	n	Number of larval instars	Unit-less	*
	t_{moult}	Duration of each larva-to-larva moult	day	*
	m_0	Neonate larval mass	mg	†
Fundamental (estimated from data)	k_0	Baseline growth rate	day^{-1}	‡
	θ	The difference between temperature sensitivities of metabolic rate and diffusion; sets the oxygen safety margin at the beginning of each instar	Unit-less	‡
	r	Development rate of the physiological processes that are triggered at the critical mass and result in moulting	day^{-1}	‡

Although parameters n and t_{moult} are principally individual-specific, we assigned them species-specific values in our analyses. The fundamental parameters are individual-specific parameters whose values were estimated from empirical data. The fundamental parameters have a major effect on the shape of the growth trajectory and are, therefore, biologically very interesting.

*Species-specific values were used in the analyses.

†Individual-specific measured values were used in the analyses.

‡Individual-specific values were estimated from the data.

the underlying mechanism remains unknown, we model it with an arbitrary function (see below).

Within an instar, the capacity of the tracheal system to supply oxygen to tissues, $D(i)$ (mol h^{-1}), is, for simplicity, assumed constant, given a constant temperature. The oxygen demand of tissues increases with growth. Hence, the oxygen demand, that is, absolute (whole-organism) metabolic rate, $R(m)$ (mol h^{-1}), will match $D(i)$ at a certain mass, and we define this mass as the critical mass (cf. Callier & Nijhout, 2011) (Note that we essentially use metabolic rate as a synonym for oxygen demand. Empirically measurable metabolic rate underestimates oxygen demand under oxygen limitation, which potentially explains some nonlinearities in empirical data on metabolic scaling, Callier & Nijhout, 2011). Whole-organism metabolic rate is assumed to scale allometrically to body mass (e.g. Glazier, 2005) with the allometric relationship being different in each instar (Sears *et al.*, 2012). Hence, whole-organism metabolic rate is given by

$$R(i,m) = c(i)m^{d(i)} \tag{3}$$

where $c(i)$ and $d(i)$ are instar-specific constants. Metabolic rate also increases with increasing temperature, and the temperature-sensitivity can be described by the Arrhenius function (Brown *et al.*, 2004) so that the whole-organism metabolic rate as a function of mass and temperature during instar i is given by

$$R(i,m,T) = c(i)m^{d(i)}e^{-\frac{f}{T}} \tag{4}$$

where f is a constant ([activation energy (kJ)]/[Boltzmann's constant (kJ K^{-1})]) and T is the absolute temperature (K). As $D(i) = R(i,m_{crit}(i),T)$ at the critical mass of instar i, we can derive the following equation for the critical mass (see Appendix A)

$$m_{\text{crit}}(i) = \sigma m_0(i)e^{\theta} \tag{5}$$

where

$$\sigma = \frac{P_{\text{crit}_{\text{early}}}}{\delta P_{\text{crit}_{\text{late}}}} \tag{6}$$

and θ is a composite parameter (see Appendix A for details) including the difference between thermal sensitivities of metabolic rate and tracheal system oxygen supply capacity (mainly diffusion). In eqn 6, $P_{\text{crit}_{\text{early}}}$ and $P_{\text{crit}_{\text{late}}}$ are critical oxygen partial pressures (kPa) early and late within an instar below which normal growth is hindered (Greenlee & Harrison, 2004a, b; Greenlee, Nebeker & Harrison, 2007) and δ is a (unit-less) parameter that defines how much maximum mass-specific tracheal conductance, $G_{\max}$ (mol g^{-1} h^{-1} kPa^{-1}), decreases during an instar ($0 < \delta < 1$) (see Appendix A; eqn A4).

The final (or peak) mass of instar i, $M(i)$, can be calculated with eqn 2 by setting $t' = t_{\text{post-crit}}$. We assume that moulting takes time t_{moult} at each larva-to-larva moult and, for simplicity, ignore loss of mass during moulting (in our study species *Pararge aegeria*, *Pararge xiphia*, *Pieris napi* and *Polyommatus icarus* [see below], average percentages of mass loss during moulting were 11, 13, 6.3 and 4.5, respectively). Thus, the final mass of instar i equals the initial mass of instar $i + 1$, and we denote the initial mass of instar i by $M(i - 1)$. Further, we define $M(0) = m_0$, where m_0 is the mass of a neonate larva. Consequently, the growth model can be expressed as the following set of equations

$$M(i) = M(i-1)e^{k(i)\left(t_{\text{crit}}(i)+\frac{t_{\text{post-crit}}}{2}\right)} \tag{7a}$$

$$M(0) = m_0 \tag{7b}$$

$$k(i) = k_0 e^{-\alpha M(i-1)} \tag{7c}$$

$$t(i) = t_{\text{crit}}(i) + t_{\text{post-crit}} + \varphi \tag{7d}$$

$$\varphi = \begin{cases} 0, \text{if } i = 1 \\ t_{\text{moult}}, \text{if } i > 1 \end{cases} \tag{7e}$$

$$t_{\text{crit}}(i) = \frac{\ln(m_{\text{crit}}(i)) - \ln(M(i-1))}{k(i)} \tag{7f}$$

$$m_{\text{crit}}(i) = \sigma M(i-1)e^{\theta} \tag{7g}$$

$$t_{\text{post-crit}} = \frac{1}{r} \tag{7h}$$

$$i = 1, \ldots, n \tag{7i}$$

where parameters k_0 and α determine the baseline level and decrease rate of $k(i)$ in relation to initial mass of an instar i, respectively, and $t(i)$ is the duration of instar i from hatching (1st instar) or beginning of moult (other instars) to the beginning of next moult (1st to the penultimate instar) or to the attainment of peak mass (final, i.e. n^{th} larval instar). Eqn 7f is derived from eqn 1 by setting $t = t_{\text{crit}}(i)$ and solving for $t_{\text{crit}}(i)$. For the final mass of an instar to be higher than the initial mass of that instar, parameter θ must satisfy

$$\theta > -\ln(\sigma) \tag{8}$$

The peak larval mass is $M(n)$, and the larval development time until the attainment of $M(n)$, t_{larva}, is

$$t_{\text{larva}} = \sum_{i=1}^{n} t(i) = n t_{\text{post-crit}} + (n-1)t_{\text{moult}} + \sum_{i=1}^{n} t_{\text{crit}}(i) \tag{9}$$

The model includes eight parameters (m_0, k_0, α, t_{moult}, σ, θ, r and n; see Table 1). Two of these (α and

σ) can be fixed for any particular analysis. Data on *M. sexta* suggest that the value of σ is ~2/3 (Greenlee & Harrison, 2005), and we use this value throughout. We fix α to 0.01, which is appropriate for our data (see Results and Supporting Information, Appendix S1). Note, however, that a single value of α is not appropriate over a wide range of body sizes as it must inherently be negatively correlated with body size (see also Supporting Information, Fig. S1). Although the predictions seem robust for the value of α used (Supporting Information, Appendix S1), it is important to pay attention to the value of α, particularly if the model is applied to a species with considerably smaller (< 95 mg) or larger (> 400 mg) body size compared to our study species. Similarly, the predictions of the model may be affected by the specific formulation of growth after the attainment of the instar-specific critical mass. We assumed a linear growth deceleration in relation to time (eqns 2 and A2). In order to evaluate the sensitivity of the predictions to this particular formulation, we repeated all the analyses (see below) with a modification of the model (eqns) 7 where growth decelerates exponentially in relation to time after the attainment of the critical mass (Appendix B).

Three parameters (m_0, t_{moult} and n) can easily be empirically measured (t_{moult} can also be inferred from growth trajectory data), and the number of instars, n, is fixed in many species. The remaining three parameters (k_0, θ and r) are biologically particularly interesting, and we call them fundamental parameters. Parameter k_0 can be interpreted as growth rate, because it has a major effect on the instar-specific growth coefficients, $k(i)$, given a constant α. Parameter θ sets the oxygen safety margin at the beginning of each instar. Therefore, given a constant σ, θ completely determines how much higher the critical mass is compared to the initial mass of each instar, critical masses always following Dyar's rule (Appendix A; Callier & Nijhout, 2011). Interestingly, critical masses follow also the temperature-size rule (Appendix A), that is, decrease with increasing temperature (Atkinson, 1994). Finally, r completely determines the time interval between the attainment of critical mass and the beginning of moult (or the attainment of the peak mass in the final instar).

We fix parameters α and σ in all analyses, and use species-specific values of t_{moult} and n to numerically estimate the individual-specific values of the fundamental parameters k_0, θ and r from empirical data (see below). The estimation of k_0, θ and r requires that larval mass is measured at a minimum of three points: at hatching (m_0), at the attainment of peak mass [$M(n)$] and at least once between these end points of the growth trajectory. Thus, the model can easily be applied to and evaluated with empirical data.

Estimating the fundamental parameters of the mechanistic model

The values of the three fundamental parameters were estimated separately for each individual by using the individual-specific growth trajectory data. First, we created a sequence of 500 equally spaced values of θ from $-\ln(\sigma) + 0.001$ (cf. inequality 8) to 3. Second, for each value of θ along the sequence, we found the values of k_0 and r that minimized the quantity $[M(n) - m_{\mathrm{peak_{obs}}})^2 + (t_{\mathrm{larva}} - t_{\mathrm{larva_{obs}}})^2]^{1/2}$ (i.e. the Euclidean distance between observed and predicted end points of a growth trajectory; $m_{\mathrm{peak_{obs}}}$ is the observed peak larval mass and $t_{\mathrm{larva_{obs}}}$ is the observed larval development time until peak larval mass) given the particular value of θ with the Nelder–Mead method (Nelder & Mead, 1965) implemented in the R function *optim* (R Core Team, 2013). Third, we derived the predicted growth trajectory for each analysed value of θ by using the particular θ value and the associated optimal values of k_0 and r. This was done up to the highest value of θ where k_0 remained positive and r remained under an upper threshold that was considered biologically realistic (we set the threshold to 2). For all values of θ where the predicted growth trajectory was derived, we calculated the residual sum of squares of that prediction when compared to the observed growth trajectory. Natural logarithms of both the observed and predicted masses were used in calculating residual sum of squares. We, however, excluded the observed moulting sizes (see below) as well as their earlier and later neighbouring observations from data when calculating residual sum of squares to avoid circularity in predicting moulting sizes, yet unavoidable autocorrelation of size over successive time points remains a potential confounding factor. That value of θ, and associated values of k_0 and r, that minimized the residual sum of squares between the observed and predicted growth trajectories were considered as the estimates for the three fundamental parameters. The R-code used in the estimation is available as Supporting Information, Appendix S3.

Dyar's rule model

We derived growth trajectories and moulting masses also with an alternative model based on Dyar's rule (R-code available in Supporting Information, Appendix S3), stating that the size increments remain the same across instars. Dyar's rule model can be derived from the mechanistic model by setting $r = \infty$ (i.e. $t_{\mathrm{post\text{-}crit}} = 0$) and $m_{\mathrm{crit}}(i) = M(i)$. We use Dyar's

rule model as a reference when comparing the predictions of the mechanistic model to Dyar's rule.

Although Dyar's rule was originally developed for linear measures of size, it can also be applied to mass as mass generally scales to the cube of linear size. We calculate size increments in mass as the ratio of instar final mass to instar initial mass (note that this is equal to the ratio of final mass of an instar to the final mass of the previous instar as we ignored loss of mass during moulting) as instar initial and final masses represent the endpoints of a single 'event' of growth. Dyar's rule would then appear as across-instar constancy of instar-wise relative mass increments. We calculated the relative mass increment, q, as

$$q = \left(\frac{m_{\text{peak}_{\text{obs}}}}{m_{0_{\text{obs}}}}\right)^{\frac{1}{n}}. \tag{10}$$

We assumed that within-instar growth is exponential and can, thus, be described with eqn 1. We also assumed that the instar-specific growth rate coefficient, $k(i)$, decreases instar by instar similarly as in the mechanistic model, and can be described by eqn 7c. The parameter k_0 must, however, be known for the application of eqn 7c in order to derive the instar-specific $k(i)$s ($i = 1, \ldots, n$). We calculated k_0 as (see Appendix C for derivation)

$$k_0 = \frac{1}{t_{\text{larva}_{\text{obs}}} - (n-1)t_{\text{moult}}} \times \sum_{i=1}^{n} \frac{\ln(M_{\text{Dyar}}(i)) - \ln(m_{0_{\text{Dyar}}}(i))}{e^{-\alpha m_{0_{\text{Dyar}}}(i)}} \tag{11}$$

where $M_{\text{Dyar}}(i)$ is the predicted final (or peak) mass of instar $i(i = 1, \ldots, n$, and $m_{0_{\text{Dyar}}}(i)$ is the predicted initial mass of instar $i(i = 1, \ldots, n; M_{\text{Dyar}}(i) = m_{0_{\text{Dyar}}}(i+1))$, with the exception of $m_{0_{\text{Dyar}}}(1) = m_{0_{\text{obs}}}$. Predicted instar durations are given by

$$t_{\text{Dyar}}(i) = \frac{\ln(M_{\text{Dyar}}(i)) - \ln(m_{0_{\text{Dyar}}}(i))}{k(i)} + \varphi \tag{12}$$

where φ is defined by the eqn 7e).

The datasets

Seven *Pararge aegeria* (L.) and seven *Pararge xiphia* (Fabricius) (Lepidoptera: Nymphalidae) larvae were weighed 19–24 and 30–35 times, respectively, from hatching to the attainment of peak mass. These larvae were random samples from laboratory populations originating from Madeira. Larvae were individually reared on the natural host plant *Dactylis glomerata* L. at 18 °C under 15 h day length.

We individually reared 15 *Pieris napi* (L.) (Lepidoptera: Pieridae) larvae on *Alliaria petiolata* (M. Bieb.) Cavara & Grande at 16 °C under 17 h day length. The larvae were random samples from laboratory populations originating from Oulu, Finland (four individuals), and Stockholm, Sweden (11 individuals). Each individual was weighed once or twice a day during the larval development.

We used data on *Polyommatus icarus* (Rottemburg) (Lepidoptera: Lycaenidae) from Leimar (1996). Five larvae were individually reared on *Medicago sativa* L. at 23 °C under 19 h day length, and weighed daily. The larvae represented a random sample from a laboratory population originating from the island of Öland.

Analyses

We estimated the parameters of the mechanistic and Dyar's rule models for each individual in all four datasets with the methods explained above. Moulting time, t_{moult}, was set to 2 days in *P. aegeria*, 3.5 days for *P. xiphia*, 1 day for *P. napi* and 0.9 days for *P. icarus* for both the mechanistic and Dyar's rule models. We derived the predicted growth trajectory, including moulting sizes, for each individual in the four datasets from each of the two growth models, and compared the predictions and observations.

Moulting masses could be unambiguously inferred for all three moults of all individuals in *P. xiphia*, for 1–3 moults in *P. aegeria* and *P. icarus* (larvae undergo three moults) individuals, and for 1–4 moults in *P. napi* (larvae undergo four moults) individuals. We defined moulting mass as the highest mass observed within an instar, followed by an observed mass decline (some mass is lost during moulting, Ayres & MacLean, 1987). We calculated relative within-instar mass increments as [highest mass observed in an instar]/[lowest mass observed in an instar] whenever moulting masses of entering and leaving the focal instar could be estimated with the above criterion.

To quantitatively assess the explanatory powers of the two growth models, we calculated the proportion of variance explained by each of the models, R^2, for each species as

$$R^2 = 1 - \frac{\sum_{j=1}^{N}(Y_j - \widehat{Y}_j)^2}{\sum_{j=1}^{N}(Y_j - \bar{Y})^2} \tag{13}$$

where N is the number of observations (observed moulting masses of all individuals of a dataset combined), Y_j is a moult mass observation j, $\widehat{Y}_j$ is the predicted moult mass for observation j, and $\bar{Y}$is the

mean of all N observations (all on a logarithmic scale).

RESULTS

Moulting sizes

Both the mechanistic and Dyar's rule models had great success in predicting the masses of moulting from one larval instar to the next in *P. aegeria*, *P. xiphia*, *P. napi* and *P. icarus* (Table 2; Fig. 1). Despite strong positive correlations between moulting masses predicted by the two models (Pearson correlations between natural logarithms of moulting masses predicted by the two models round to 1.0 in each species), the predictions of the mechanistic model were systematically slightly higher than those of the Dyar's rule model (the difference ln[mechanistic prediction] – ln[Dyar's rule prediction]: *P. aegeria*, mean = 0.0735, $t_{20} = 2.75$, $P = 0.012$; *P. xiphia*, mean = 0.0487, $t_{20} = 4.33$, $P = 0.00032$; *P. napi*, mean = 0.271, $t_{59} = 9.80$, $P < 0.0001$; *P. icarus*, mean = 0.122, $t_{14} = 6.61$, $P < 0.0001$). Moreover, the difference between the predictions of the two models increases moult by moult as indicated by significant regression slopes in relation to moult number in linear mixed-effects model analyses conducted with the R function *lme* (Pinheiro *et al.*, 2013) with individual identity as a random effect (regression slopes derived with the restricted maximum likelihood method: *P. aegeria*, estimate = 0.0254, $t_{13} = 2.22$, $P = 0.045$; *P. xiphia*, estimate = 0.0150, $t_{13} = 3.23$, $P = 0.0065$; *P. napi*, estimate = 0.0742, $t_{44} = 7.86$, $P < 0.0001$; *P. icarus*, estimate = 0.0486, $t_9 = 6.19$, $P = 0.0002$). Both models predicted moulting masses well, even in cases where there was some mismatch between the observed and predicted timing of moulting (Supporting Information, Figs S2, S5), further suggesting that a size-dependent mechanism underlies moulting.

Table 2. Proportions of variance in moulting mass (on a logarithmic scale) that the mechanistic and Dyar's rule models explained in the four studied species (i.e. $R^2 \times 100\%$)

	Proportion of explained variance in moulting mass data (%)	
Species	Mechanistic model	Dyar's rule model
Pararge aegeria	98.2	99.0
Pararge xiphia	99.4	99.2
Pieris napi	96.1	96.0
Polyommatus icarus	98.4	98.6

The results were not sensitive to the value of parameter α used in deriving the predictions (Supporting Information, Appendix S1) or to the function describing growth after the attainment of the critical mass (Appendix B), indicating that our results are robust and not artefactual consequences of chosen parameter values or functional forms.

Within-instar mass increments

The mechanistic model predicts decreasing mass increments (growth ratios) across instars (Fig. 2). The scarce observations on within-instar mass increments were largely consistent with the predictions of the mechanistic model [Fig. 2; Pearson correlations between observations and predictions of the mechanistic model: *P. aegeria*, correlation coefficient $r_P = 0.284$ (95% CI: −0.228 to 0.673), d.f. = 15, $P = 0.27$; *P. xiphia*, $r_P = 0.503$ (0.160 to 0.738), d.f. = 26, $P = 0.0064$; *P. napi*, $r_P = 0.293$ (0.035 to 0.514), d.f. = 55, $P = 0.027$; *P. icarus*, $r_P = 0.419$ (−0.340 to 0.847), d.f. = 7, $P = 0.26$), with the notable exception of the first instar in *P. napi* that had a much higher mass increment (Fig. 2).

DISCUSSION

The mechanistic growth model developed here successfully explained moulting masses of four lepidopteran species with different numbers of larval instars and representing three different butterfly families (Nymphalidae, Pieridae, Lycenidae). Overall, the high explanatory power of the mechanistic model is in accordance with the ODIM hypothesis (Callier & Nijhout, 2011), and indirectly supports the presence of a size-dependent mechanism underlying moult induction. The mechanistic model also predicts decreasing critical mass with increasing temperature (Appendix A), which is in line with some empirical evidence (Ghosh, Testa & Shingleton, 2013; but see Davidowitz, D'Amico & Nijhout, 2004). Finally, the mechanistic model predicts increasing deviations from Dyar's rule moult by moult but, within the range of the experimental data, the deviations were generally small, suggesting that the ODIM mechanism potentially explains Dyar's rule – as well as the most common deviation from it.

Implications of the oxygen-limitation mechanism

It is worth noting that our model derivation found three important parameters (k_0, θ and r) that correspond to the three fundamental parameters that were derived by Nijhout *et al.* (2006) and Nijhout,

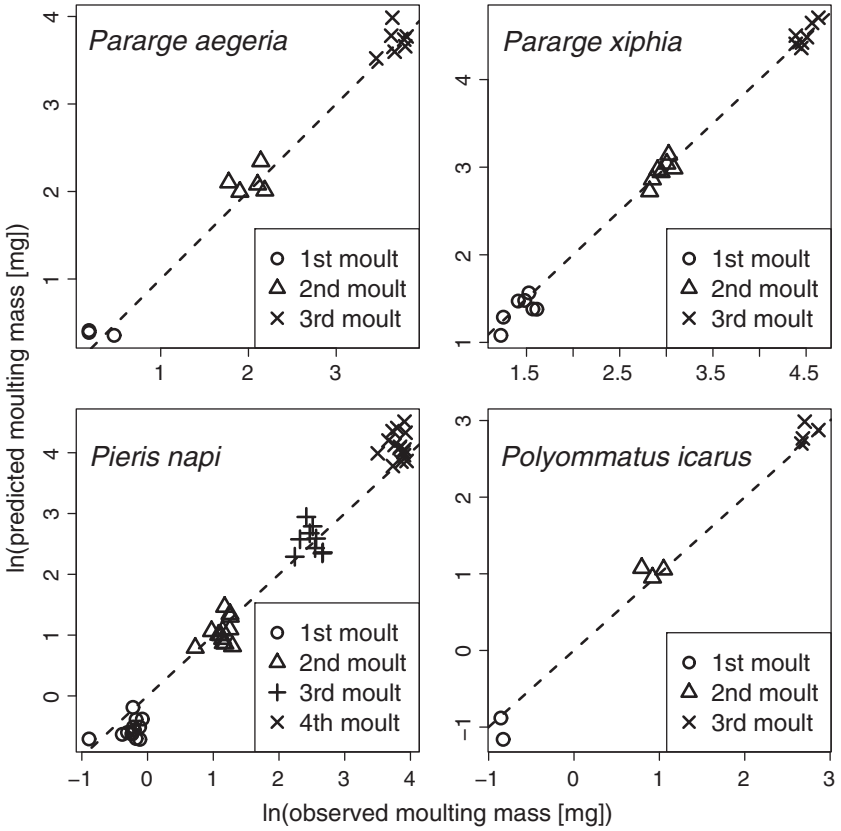

Figure 1. Predicted moulting masses for larva-to-larva moults derived from the mechanistic growth model plotted against observed moulting masses (both on a logarithmic scale) for *Pararge aegeria*, *Pararge xiphia*, *Pieris napi* and *Polyommatus icarus*. Dashed lines are 1:1 reference lines, indicating perfect match between predictions and observations. There are 15, 21, 50 and 9 data points in *P. aegeria*, *P. xiphia*, *P. napi* and *P. icarus*, respectively.

Roff & Davidowitz (2010) for the final instar *M. sexta*, that is, for the species from which the empirical evidence supporting the ODIM hypothesis comes (Callier & Nijhout, 2011). The fundamental parameters *sensu* Nijhout *et al.* (2006, 2010) are growth rate, the critical mass where the physiological preparation for pupation begins, and the interval to cessation of growth (ICG; time from the attainment of the critical mass to the beginning of moulting). In our model, k_0 corresponds to the growth rate, θ defines the critical mass (the critical mass scales to e^{θ}), and $1/r$ corresponds to ICG. However, we emphasize that although the interpretation of the three fundamental parameters in our model corresponds to the findings of Nijhout *et al.* (2006, 2010), they are not completely identical, but identify the same key factors affecting growth and development. The alternative definitions of critical mass may further confound the issue. Traditionally critical mass has been defined as the mass at which the time course of pupation becomes independent of nutrition (Nijhout & Williams, 1974; Nijhout *et al.*, 2006; Callier & Nijhout, 2013; e.g. Callier *et al.*, 2013 for a modification of this for *Drosophila melanogaster*), whereas the ODIM hypothesis defines critical mass as the mass at which oxygen demand meets the oxygen supply capacity (Callier & Nijhout, 2011). Critical mass is essentially an individual-level parameter but, due to methodological constraints, it has been measured empirically only at the population-level so far. An advantage of our mechanistic growth model is that the fundamental parameters, including critical mass, can be estimated individually. Even though the estimated values of the fundamental parameters were quite consistent within species (Table S3), there was intraspecific variation, which is unsurprising given that there was also intraspecific variation in development time and peak larval mass. This emphasizes the need for individual-wise parameter estimation. Owing to the above reasons, the values of the three fundamental growth parameters estimated with our model may not be equal to those estimated using the methodology of Nijhout *et al.* (2006) if applied to the same data.

Oxygen-dependent induction of moulting would mean that changing oxygen partial pressure changes

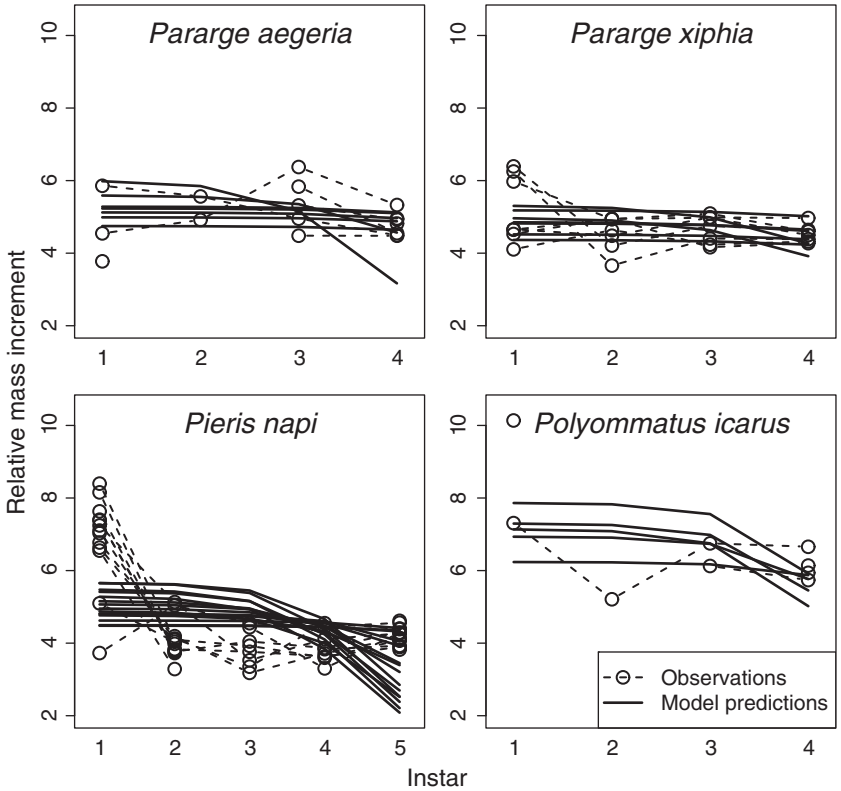

Figure 2. Observed (open circles) and predicted (thick continuous lines) instar-specific relative mass increments ([instar final mass]/[instar initial mass]) for the four studied species. Observations and predictions concerning a particular individual are connected with a line (dashed lines for observations). Observations are sparse because not all moulting masses of an individual could be rigorously determined in each case. Predictions were derived with the mechanistic model. Dyar's rule model predicts that the size increment remains constant over instars, which would appear as horizontal lines in the figure. Note that the scale of the vertical axis is the same in all panels.

critical mass. Decreasing oxygen partial pressure (hypoxia) does indeed decrease critical mass (Callier & Nijhout, 2011; Callier *et al.*, 2013) and body size (VandenBrooks *et al.*, 2012; Harrison *et al.*, 2013; reviewed by Harrison *et al.*, 2006; Harrison, Kaiser & VandenBrooks, 2010). Our model predicts exactly the same response to hypoxia. Because hypoxia decreases the oxygen safety margin at the beginning of an instar, the value of θ should decrease under hypoxia (note that θ sets the oxygen safety margin at the beginning of an instar). Decreasing θ results in decreasing instar-specific critical mass (cf. eqn 7g). Provided that r (i.e. ICG) is rather constant, decreasing θ results also in decreasing final mass of an instar, and this effect would be enhanced by decreasing k_0, which would be expected as growth rate generally decreases under hypoxia (Harrison *et al.*, 2006, 2013; VandenBrooks *et al.*, 2012; Callier *et al.*, 2013). Increasing oxygen partial pressure (hyperoxia) is predicted to have the opposite effect; instar-specific critical and possibly final masses should increase as the oxygen safety margin at the beginning of an instar (θ) increases, and growth rate (k_0) may (Frazier, Woods & Harrison, 2001; Harrison *et al.*, 2013) or may not increase (Harrison *et al.*, 2006, 2013; VandenBrooks *et al.*, 2012; Callier *et al.*, 2013). In reality, hyperoxia may slightly increase body mass (Frazier *et al.*, 2001; Harrison *et al.*, 2010; VandenBrooks *et al.*, 2012) or moulting mass (Callier & Nijhout, 2011), but this response has only been found in some species studied and at some oxygen partial pressures. A strong prediction of our mechanistic model is that hyperoxia increases the instar-specific critical mass, but this response was not found in the only study testing the prediction so far (Callier *et al.*, 2013). Possibly, hyperoxia does not increase critical mass if the tested oxygen concentrations induced oxidative stress or other detrimental effects. Alternatively, other rate-limiting metabolic processes that are not affected by oxygen partial pressure become more important than oxygen availability under hyperoxia (cf. Callier & Nijhout, 2014). It appears that more physiological work on several species is needed to assess the alternatives rigorously.

The effect of oxygen partial pressure (particularly hypoxia) on body mass has mainly been reported in holometabolous insects (Harrison *et al.*, 2006, 2010), whereas evidence for an effect of oxygen partial pressure on body size is scarce and controversial for hemimetabolous insects (Harrison *et al.*, 2006; VandenBrooks *et al.*, 2012). Moreover, nymphs of hemimetabolous insects may compensate for the limited oxygen supply attainable via diffusion by increasing active ventilation of the tracheal system towards the end of each instar (Greenlee & Harrison, 2004a). Active ventilation may make hemimetabolous nymphs less vulnerable to oxygen depletion than holometabolous larvae, among which active ventilation has been reported only under extremely low oxygen partial pressure (Greenlee *et al.*, 2013). It is also known that moulting is induced by abdominal stretch receptors in some hemimetabolous insects (reviewed by Callier & Nijhout, 2013), which makes sense especially in ectoparasitic blood-feeding species where moult can take place after a sufficient blood meal. Owing to these reasons, the ODIM hypothesis (Callier & Nijhout, 2011) underlying our mechanistic model seems more applicable to holometabolous than hemimetabolous insects.

Within-instar mass increments

The mechanistic model predicts decreasing mass increments (growth ratios) across instars in our data sets (Fig. 2). Despite this, the predictions were close to Dyar's rule in most cases as indicated by both models' excellent fit to observations (Table 2), suggesting that an empirical demonstration of a deviation from Dyar's rule is difficult in many cases. The critical masses were assumed to follow Dyar's rule in the mechanistic model (Appendix A), so the deviations from Dyar's rule arise due to growth after the attainment of the critical mass. Increasing deviations from Dyar's rule are predicted when the predicted time interval between the attainment of the critical mass and moult initiation ($1/r$, i.e. ICG) becomes longer. Lengthening ICG is associated with decreasing critical mass, and there is a positive linear relationship between the parameter θ (i.e. the determinant of the critical mass) and the predicted reduction in instar-specific relative mass increments ([final instar relative mass increment]/[first instar relative mass increment]), as illustrated by our largest data set (*P. napi*) where the strongest deviations from Dyar's rule are predicted (Pearson correlation: $r_P = 0.974$ [95% CI: 0.922–0.992], d.f. = 13, $P < 0.001$). Conversely, it is evident that the oxygen-dependent moult induction mechanism provides a plausible explanation for the empirically derived Dyar's rule within biologically realistic parameter space. It is important to note that this explanation is not mutually exclusive to the 'investment principle' (i.e. optimization of resource acquisition rate) by Hutchinson *et al.* (1997). In fact, the central assumption of the 'investment principle' that resource acquisition rate depends on structures that only grow at moult holds also for insect respiration, because the tracheal system principally grows only at moult and oxygen is the rate-limiting resource.

Despite parallel predictions by our mechanistic model and Hutchinson *et al.*'s (1997) 'investment principle' model, these two models are quite different. The 'investment principle' model was an optimization model where timing of moults was adjusted to maximise growth rate across the larval stage, growth rate remaining constant within an instar, and depending only on instar initial mass. In our model, moulting times are not optimized but they are mechanistic consequences of tracheal oxygen supply, tissue oxygen demand and rate of the processes required to initiate moulting. We also modelled growth differently as within-instar growth rate was assumed to be proportional to mass before the attainment of the critical mass, and decelerating thereafter, which seems realistic for insect larvae (Nylin *et al.*, 1996; Esperk & Tammaru, 2004; Nijhout *et al.*, 2006; Grunert *et al.*, 2015). The growth coefficients were instar-specific in our model as indicated by empirical evidence (Nijhout *et al.*, 2006), and depended on the initial mass of an instar.

The observations were mainly consistent with Dyar's rule, with the notable exception of the first instar in *P. napi* that had a much higher mass increment (Fig. 2). This deviation is not exceptional (cf. Hutchinson *et al.*, 1997), and may be explained by a significant amount of cutaneous respiration in the first instar *P. napi*, which would raise the critical mass above the point determined by the tracheal system alone. Cutaneous respiration would affect growth more in *P. napi* than in *Pararge* spp. because neonate *P. napi* larvae are much smaller than those of *Pararge* spp. This results in a higher surface area to volume ratio in *P. napi*, facilitating extensive cutaneous respiration. Accordingly, the evolution of some extraordinary growth patterns, such as the enormous mass increments of the final instar *Maculinea* spp. butterflies (Elmes *et al.*, 2001) or the pronounced instar-wise increase in the relative mass increment in *M. sexta* (Grunert *et al.*, 2015) may require special respiratory mechanisms or adaptations, like exaggerated size of the tracheal system.

Predictions concerning growth trajectories

In addition to moulting, the mechanistic and Dyar's rule models had high success in predicting the

overall growth trajectories, the proportions of explained variance in growth trajectory data in the four study species ranging from 94.1% to 98.9% (Table S2). However, in order to derive even better-fitting and more realistic predictions for the overall growth trajectories, the potentially mass-dependent deceleration of growth after the attainment of the critical mass needs to be modelled in a more realistic way. This need is indicated by the very short estimates for ICGs for some individuals (<18 h; cf. Table S3), which are likely underestimates, and by mismatching predictions for the final instar larvae in many cases (Supporting Information, Figs S2–S6). In addition, the effect of potential cutaneous respiration in very small larvae (see above) needs to be included in the model. Modelling these aspects of larval growth more realistically requires more physiological data, calling for novel experiments on larval growth.

CONCLUSIONS

Although the endocrinological cascade resulting in moulting is relatively well understood in the model organisms *M. sexta* and *D. melanogaster* (Shingleton, 2011; Callier & Nijhout, 2013; Nijhout *et al.*, 2014), the exact trigger of that cascade (i.e. the size-sensing mechanism) remains unknown. One plausible candidate is the ODIM hypothesis (Callier & Nijhout, 2011; see above), which states that moulting is triggered by oxygen stress as the larva grows larger. This was an underlying assumption of our mechanistic model that successfully predicted moulting size data in four butterflies. Moreover, ODIM provides a candidate mechanism for the empirically described pattern of Dyar's rule and for the most common deviation (monotonic decrease in the growth ratio) from it.

Juvenile growth and development determine a holometabolous insect's age and size at maturity as well as the initial amount of reserves available for adult reproduction. Thus, understanding the mechanistic basis of developmental decision-making, such as moulting, is of fundamental importance for understanding life history evolution (cf. Flatt & Heyland, 2011). The successful modelling of moulting decisions presented here, therefore, opens a way for developing better life history models for insects. Mechanistic models are required for the analysis of evolutionary constraints on growth and moulting strategies, as well as phenotypic plasticity in them. For instance, the ODIM hypothesis has important implications for phenotypic plasticity in central life history traits as the predicted temperature-size response of critical mass may help in understanding the evolution and emergence of the temperature-size rule (Atkinson, 1994). Further work is required for a rigorous assessment of this issue and the model developed here can be a helpful tool in such research.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1. Parameter α: species-specific values and a sensitivity analysis.
Appendix S2. Supplementary figures and tables.
Appendix S3. R-code of the model.

SHARED DATA

Data deposited in the Dryad digital repository (Kivelä *et al.*, 2015).

APPENDIX A

Details of derivation of the mechanistic model

Instantaneous growth rate before the attainment of the critical mass in instar i is given by

$$\frac{\mathrm{d}m}{\mathrm{d}t} = k(i)m \qquad (A1)$$

where $k(i)$ is an instar-specific constant ($i = 1, \ldots, n$; n is the number of instars) and m is mass. For simplicity, we assume that instantaneous growth rate decreases linearly with respect to time during the growth period between the attainment of the instar-specific critical mass and the beginning of the next moult (or at the attainment of the peak larval mass at the final larval instar). This formulation means that growth ceases just before the beginning of a moult (or at the attainment of peak larval mass at the final instar). Hence, instantaneous growth rate after the attainment of the critical mass is given by

$$\frac{\mathrm{d}m}{\mathrm{d}t'} = k(i)m\left(1 - \frac{t'}{t_{\text{post-crit}}}\right), \qquad (A2)$$

where $t_{\text{post-crit}}$ is the time from the attainment of critical mass to moulting (or to the attainment of the peak larval mass at the final instar). Integrating eqn A1 with the initial condition $m(0) = m_0(i)$, where $m_0(i)$ is the initial mass of instar i, gives eqn 1, and integrating eqn A2 with the initial condition $m(0) = m_{\text{crit}}(i) = m_0 e^{k(i)t\text{crit}(i)}$ gives eqn 2. The following procedure for deriving eqn (5) for the critical mass can be applied to any instar, so we ignore the instar index i in the following derivations to simplify notation.

The maximum mass-specific tracheal conductance, $G_{\max}$ (mol g^{-1} h^{-1} kPa^{-1}) is defined as

$$G_{\max} = \frac{B}{P_{\text{crit}}}, \qquad (A3)$$

where B is mass-specific metabolic rate (mol g^{-1} h^{-1}) and P_{crit} is the critical oxygen partial pressure (kPa) below which normal growth is hindered (Greenlee & Harrison, 2004a, b; Greenlee *et al.*, 2007). We emphasize that, in our model, P_{crit} refers to critical oxygen partial pressure for growth, which is higher than the critical oxygen partial pressure for metabolic rate (Harrison *et al.*, 2014), as would be expected under the ODIM hypothesis. Furthermore, P_{crit} is independent of size across species (Harrison *et al.*, 2014), supporting our formulation. $G_{\max}$ decreases during growth within each instar (Greenlee & Harrison, 2005). Hence, we define

$$G_{\max_{late}} = \delta G_{\max_{early}}, \qquad (A4)$$

where $G_{\max_{late}}$ and $G_{\max_{early}}$ refer to the maximum mass-specific tracheal conductances at the end and at the beginning of an instar, respectively, δ being a

(unit-less) parameter that defines how much $G_{\max}$ decreases during an instar ($0 < \delta < 1$). The above definitions imply

$$D = G_{\max} m P_{\text{crit}} \tag{A5}$$

Solving eqn A5 for $G_{\max}$ gives

$$G_{\max} = \frac{D}{m P_{\text{crit}}}. \tag{A6}$$

Substituting eqn A6 into eqn A4, and noting that $D = R(m_{\text{crit}}, T)$, eqn A4 can be rewritten as

$$\frac{R(m_{\text{crit}}, T)}{m_{\text{crit}} P_{\text{crit}_{\text{late}}}} = \frac{\delta D}{m_0 P_{\text{crit}_{\text{early}}}}, \tag{A7}$$

where $P_{\text{crit}_{\text{early}}}$ and $P_{\text{crit}_{\text{late}}}$ ($P_{\text{crit}_{\text{early}}} < P_{\text{crit}_{\text{late}}}$) are the early- and late-instar critical oxygen partial pressures, respectively (see Greenlee & Harrison, 2005). Solving D from eqn A7 gives

$$D = \frac{c P_{\text{crit}_{\text{early}}} m_0 m_{\text{crit}}^{d-1} e^{-\frac{\gamma}{T}}}{\delta P_{\text{crit}_{\text{late}}}}. \tag{A8}$$

Note that we have replaced parameter f in R (m_{crit}, T) with γ in eqn A8 because temperature-sensitivity of D is mainly affected by diffusion, and has therefore different temperature-sensitivity from that of metabolic rate. Metabolic rate is more sensitive to temperature than diffusion, meaning that $f > \gamma$, and resulting in the temperature-size rule for critical mass (i.e. decreasing critical mass with increasing temperature; see below). Eqn A8 can be simplified as

$$D = c\sigma m_0 m_{\text{crit}}^{d-1} e^{-\frac{\gamma}{T}}, \tag{A9}$$

where

$$\sigma = \frac{P_{\text{crit}_{\text{early}}}}{\delta P_{\text{crit}_{\text{late}}}} \tag{A10}$$

The critical mass is a constant multiple, s, of the initial mass of an instar (Callier & Nijhout, 2011). Thus, we can rewrite eqn A9 as

$$D = c\sigma m_0 (s m_0)^{d-1} e^{-\frac{\gamma}{T}} = c\sigma m_0^d s^{d-1} e^{-\frac{\gamma}{T}}. \tag{A11}$$

Now, we can calculate the critical mass by setting D (eqn A11) equal to $R(m_{\text{crit}}, T) = c m_{\text{crit}}^d e^{-f/T}$ (cf. eqn 4), and solving the equation for m_{crit}, giving

$$m_{\text{crit}} = \sigma^{\frac{1}{d}} m_0 s^{1-\frac{1}{d}} e^{\frac{f-\gamma}{dT}}. \tag{A12}$$

Due to different temperature sensitivities of diffusion and metabolic rate, the ratio m_{crit}/m_0 is exactly s only at a particular temperature. Setting this temperature as a reference temperature, T_{ref}, and using eqn A12 for critical mass gives

$$s = \sigma e^{\frac{f-\gamma}{T_{\text{ref}}}}. \tag{A13}$$

Substituting this into eqn A12 gives

$$m_{\text{crit}} = \sigma m_0 e^{\frac{d(f-\gamma)+\gamma-f}{dT_{\text{ref}}} + \frac{f-\gamma}{dT}}. \tag{A14}$$

We define $\Delta = f - \gamma$. Consequently, critical mass can be expressed as

$$m_{\text{crit}} = \sigma m_0 e^{\frac{\Delta(T(d-1)+T_{\text{ref}})}{dTT_{\text{ref}}}}. \tag{A15}$$

Eqn A15 implies temperature-size rule in critical mass as differentiating this equation with respect to temperature, T, gives

$$\frac{\partial m_{\text{crit}}}{\partial T} = -\frac{\Delta\sigma m_0}{dT^2} exp\left(\frac{\Delta(T(d-1)+T_{\text{ref}})}{dTT_{\text{ref}}}\right). \tag{A16}$$

In eqn A16, the coefficient $-\Delta\sigma m_0/dT^2$ is always negative, while the exponential part is always positive, which means that the derivative $\partial m_{\text{crit}}/\partial T$ is always negative, that is, the critical mass always decreases with increasing temperature.

By setting $T = T_{\text{ref}}$, eqn A15 simplifies to

$$m_{\text{crit}} = \sigma m_0 exp\left(\frac{\Delta}{T_{\text{ref}}}\right) \tag{A17}$$

which can be further simplified by setting $\Delta/T_{\text{ref}} = \theta$, giving

$$m_{\text{crit}} = \sigma m_0 e^{\theta} \tag{A18}$$

which is eqn 5 used in the growth model.

APPENDIX B

Alternative model for post-critical-mass growth

Here, we assume that instantaneous growth rate decelerates exponentially with respect to time after the attainment of the critical mass within each larval instar. Consequently, instantaneous growth rate after the attainment of the critical mass can be expressed as

$$\frac{dm}{dt'} = k(i) m e^{-\beta t'} \tag{B1}$$

where β is a parameter determining the deceleration rate of instantaneous growth rate in relation to time.

Integrating eqn B1 with the initial condition $m(0) = m_{\text{crit}}(i)$ gives

$$m_{\text{post-crit}}(i,t') = \frac{m_{\text{crit}}(i)(k(i)e^{-\beta t'} - \beta - k(i))}{-\beta} \qquad (B2)$$

where $m_{\text{post-crit}}(i,t')$ is mass of instar i at time t', and $0 \le t' \le t_{\text{post-crit}}$. Other components of the model remain constant, so setting $t' = t_{\text{post-crit}}$ and substituting $m_0 e^{k(i)t_{\text{crit}}(i)}$ for $m_{\text{crit}}(i)$ in eqn B2 results in the following growth model

$$M(i) = \frac{M(i-1)e^{k(i)t_{\text{crit}}(i)}(k(i)e^{-\beta t_{\text{post-crit}}} - \beta - k(i))}{-\beta} \qquad (B3a)$$

$$M(0) = m_0 \qquad (B3b)$$

$$k(i) = k_0 e^{-\alpha M(i-1)} \qquad (B3c)$$

$$t(i) = t_{\text{crit}}(i) + t_{\text{post-crit}} + \varphi \qquad (B3d)$$

$$\varphi = \begin{cases} 0, \text{if } i = 1 \\ t_{\text{moult}}, \text{if } i > 1 \end{cases} \qquad (B3e)$$

$$t_{\text{crit}}(i) = \frac{\ln(m_{\text{crit}}(i)) - \ln(M(i-1))}{k(i)} \qquad (B3f)$$

$$m_{\text{crit}}(i) = \sigma M(i-1)e^{\theta} \qquad (B3g)$$

$$t_{\text{post-crit}} = \frac{1}{r} \qquad (B3h)$$

$$i = 1, \ldots, n. \qquad (B3i)$$

The model B3 contains one extra parameter, β, compared with the model 7. In order to derive an estimate of the individual-specific value of β, we modified the parameter estimation procedure tailored for model 7 so that the Nelder–Mead method (Nelder & Mead, 1965) was used for simultaneous optimization of k_0, r and β, given a value of θ. Moreover, we excluded those values of θ where the estimate of β was negative from the minimization of residual sum of squares. Otherwise, the procedure was similar to that used for model 7.

The moulting masses predicted by the models 7 and B3 were practically identical as indicated by the effectively perfect correlations between the natural logarithms of predicted moulting masses of these two models in all of the four study species (all Pearson correlation coefficients rounded to 1.00).

APPENDIX C

Derivation of eqn 11

The duration of growth during the larval stage (i.e. larval development time until peak mass, excluding time spent in moulting) can be written as

$$t_{\text{larva}_{\text{obs}}} - (n-1)t_{\text{moult}} = \sum_{i=1}^{n} \frac{\ln(M_{\text{Dyar}}(i)) - \ln(m_{0_{\text{Dyar}}}(i))}{k(i)} \qquad (C1)$$

where $M_{\text{Dyar}}(i)$ is the predicted final (or peak) mass of instar i (i = 1, …, n), and $m_{0_{\text{Dyar}}}(i)$ is the predicted initial mass of instar i ($i = 1, \ldots, n; M_{Dyar}(i) = m_{0_{\text{Dyar}}}(i+1)$), with the exception of $m_{0_{\text{Dyar}}}(1) = m_{0_{\text{obs}}}$. The growth rate coefficient of instar i, $k(i)$, is given by eqn 7c, which can be substituted into eqn C1 giving

$$t_{\text{larva}_{\text{obs}}} - (n-1)t_{\text{molt}} = \sum_{i=1}^{n} \frac{\ln(M_{\text{Dyar}}(i)) - \ln(m_{0_{\text{Dyar}}}(i))}{k_0 e^{-\alpha m_{0_{\text{Dyar}}}(i)}}. \qquad (C2)$$

Eqn C2 can be rewritten as

$$t_{\text{larva}_{\text{obs}}} - (n-1)t_{\text{molt}} = \frac{1}{k_0} \sum_{i=1}^{n} \frac{\ln(M_{\text{Dyar}}(i)) - \ln(m_{0_{\text{Dyar}}}(i))}{e^{-\alpha m_{0_{\text{Dyar}}}(i)}} \qquad (C3)$$

from which it is straightforward to solve k_0, giving eqn 11.